

E. SAM CLELAND

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EDUCATION

Fordham University, Gabelli School of Business | MBA, Financial Technology (GPA: 3.94)

Aug 2024 – May 2026

- Relevant Coursework: Cryptocurrency Algorithmic Trading, NLP, RAG & Context Engineering, Data Analytics for Finance, Intro to Fintech & Portfolio Management, Web Analytics
- Graduate Research Assistant: Image processing project using Multimodal LLMs and CNNs for facial feature extraction (symmetry, beauty scores)
- VP of Tech & Research, Fordham GSB AI Society: Manage club website and develop AI workshops and events

McGill University | BEng, Computer Engineering

Aug 2015 – Dec 2019

- Relevant Coursework: Applied Machine Learning, Artificial Intelligence, Intro to Software Engineering, Engineering Economy
- Head Teaching Assistant for Design Principles & Methods (core ECE course)

EXPERIENCE

MNNC Group x Fordham | AI Research Collaborator

2025 – Present

- Designed and executed experiments to extract alpha signals from LLMs (Gemini, OpenAI GPT, xAI Grok), systematically comparing model outputs as predictive signals for cryptocurrency allocation strategies
- Built end-to-end backtesting pipelines in Python integrating minute-level price data (Binance API) with LLM-generated signals to evaluate risk-adjusted returns across multiple cryptocurrencies
- Iterated on prompt engineering and structured output parsing to reliably extract tradable signals from model responses at scale

Trinity Group International | Analyst / Technical Support & E-Commerce Seller

Mar 2020 – Jun 2024

Amazon/Walmart E-Commerce

- Built automated research pipelines to evaluate product listings, detect buy-box shifts and new competitor entries, and surface actionable insights — directly analogous to feature engineering for systematic alpha research
- Automated profitability audits using Python and standardized reporting across accounts; integrated external data tools (Helium 10, Keepa, Amazon Alerts) into systematic workflows
- Created scalable automation and new tooling workflows adopted across teams, demonstrating ability to rapidly prototype and deploy research infrastructure

NorthPage Inc. | Software Engineering Intern (R&D)

May – Aug 2018

AI Digital Marketing Company

- Participated in R&D group serving major clients including IBM; platform supported 2,000+ concurrent users
- Completed 30+ development tasks in 4 months including code refactoring, frontend improvements, and regex-based data consolidation

RESEARCH & ACADEMIC PROJECTS

LLM-Based Crypto Sentiment Trading Signal (Cryptocurrency Algorithmic Trading) | Fordham

- Engineered a multi-asset trading signal using Gemini API to classify market sentiment (bearish/neutral/bullish) as numerical scores fed into a regression model; backtested across Jan 2023–Sep 2025 using minute-level Binance data
- Directly aligns with systematic alpha development and AI-augmented quant research workflows

Meal-Prep AI Service (RAG & Context Engineering) | Fordham

- Built full-stack RAG system (React/FastAPI/ChromaDB) using GPT-4o and Gemini for personalized meal planning with Google Calendar integration; deployed on Vercel — demonstrates agentic workflow and LLM pipeline design

Contract Risk Analysis NLP Pipeline (NLP) | Fordham

- Automated clause-level risk classification (low/moderate/high) using NLTK, WordNet, and a Bidirectional LSTM model in TensorFlow — demonstrates production-grade NLP pipeline construction

COMPETITIONS

Fordham / Deloitte March Madness Data Competition

Feb – Apr 2025

- Finished 2nd of 10 teams; built a TensorFlow Siamese Neural Network to predict game win probabilities using tournament data enriched with external CBB stats and campus altitude data — demonstrates rigorous model validation and creative feature engineering

United Nations Technology Sprint — UNMAS/OICT

Dec 2025

- Delivered a production web application for UNMAS to replace manual Excel-based procedures; led SharePoint List architecture design, PowerApps reports integration, and Power BI dashboard for annual UN website publication

TECHNICAL SKILLS

Languages & Frameworks: Python, JavaScript/TypeScript, Java, C, C#, SQL, React, Next.js, Java Spring Boot

ML / AI Tools: TensorFlow, PyTorch, NLTK, Claude Code, GPT-based agents, Gemini API, OpenAI API, xAI API, LangChain, ChromaDB, RAG pipelines

Quant & Data: Backtesting frameworks, Binance API, algorithmic trading signal development, feature engineering, regression modeling, time-series analysis, Hadoop

Infrastructure: Linux, Git, Vercel, FastAPI, SharePoint/PowerApps, PowerAutomate, Power BI